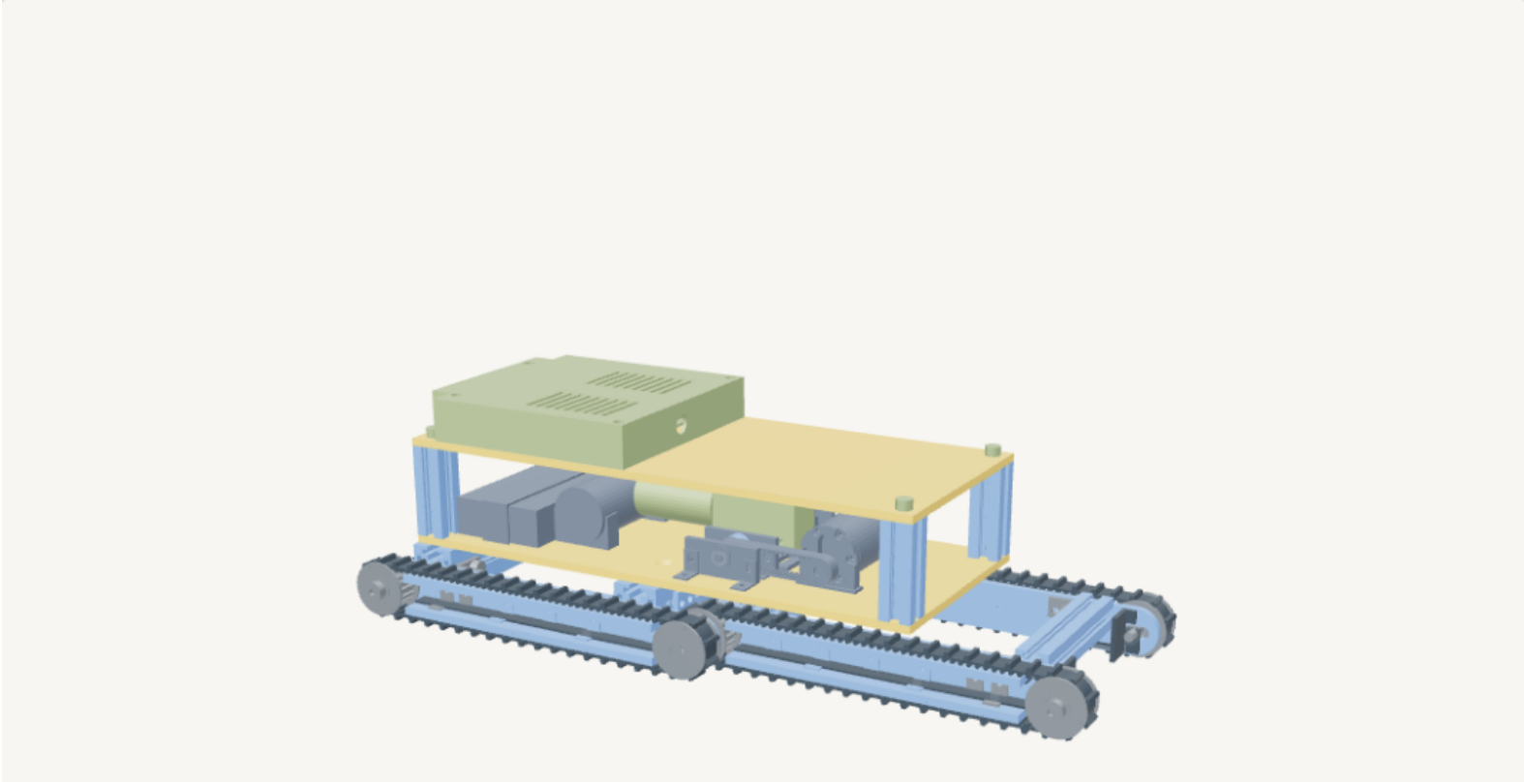
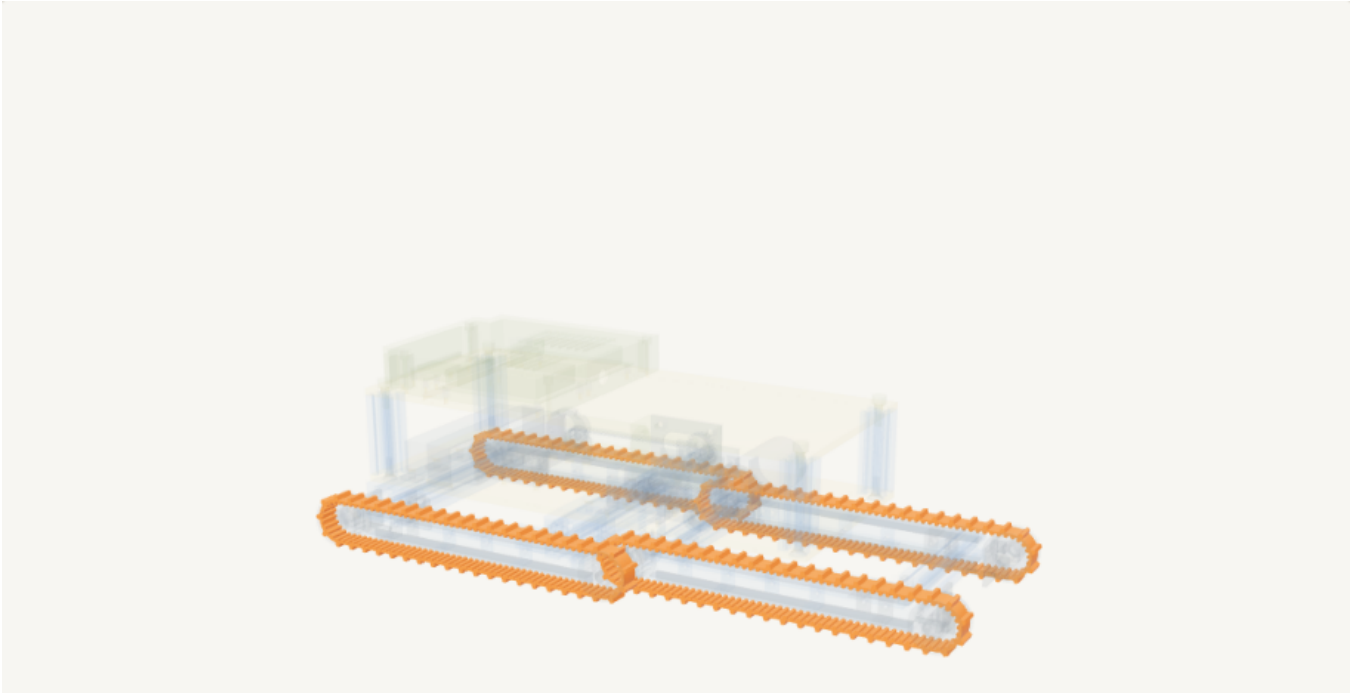
Parts catalogue: the track unit on Nhien's Rev B frame

One page per part. Orange = that part in the shared model of Nhien's Rev B assembly. Blue lines are links: click the vendor line to open the product page. Rev B sheet numbers refer to 'RB4107 Fabrication drawings (Rev B).pdf'.



Track, two options (next two pages):
A my printed TPU track x4, or
B Nhien's belt x4 + 88 bonded cleats.
Pulleys x8 bought plain bore, either way.
Nhien's: plates, rails, shafts, bearings, shoulder screws, bars, deck, motors, electronics (his BOM).

Option A: Track, printed



Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Quantity: 4

MAKE: 3D print (mine)

What it does

The printed TPU belt: 110 teeth inside on the toothed wheels, 45 cleats 3 mm high outside, spaced so five span one ES1 step. Fits Rev B's pulleys and 225 mm spacing; its cleat tips sit 5 mm under the deck. The other version of this pack uses Nhien's stock belt with bonded cleats instead.

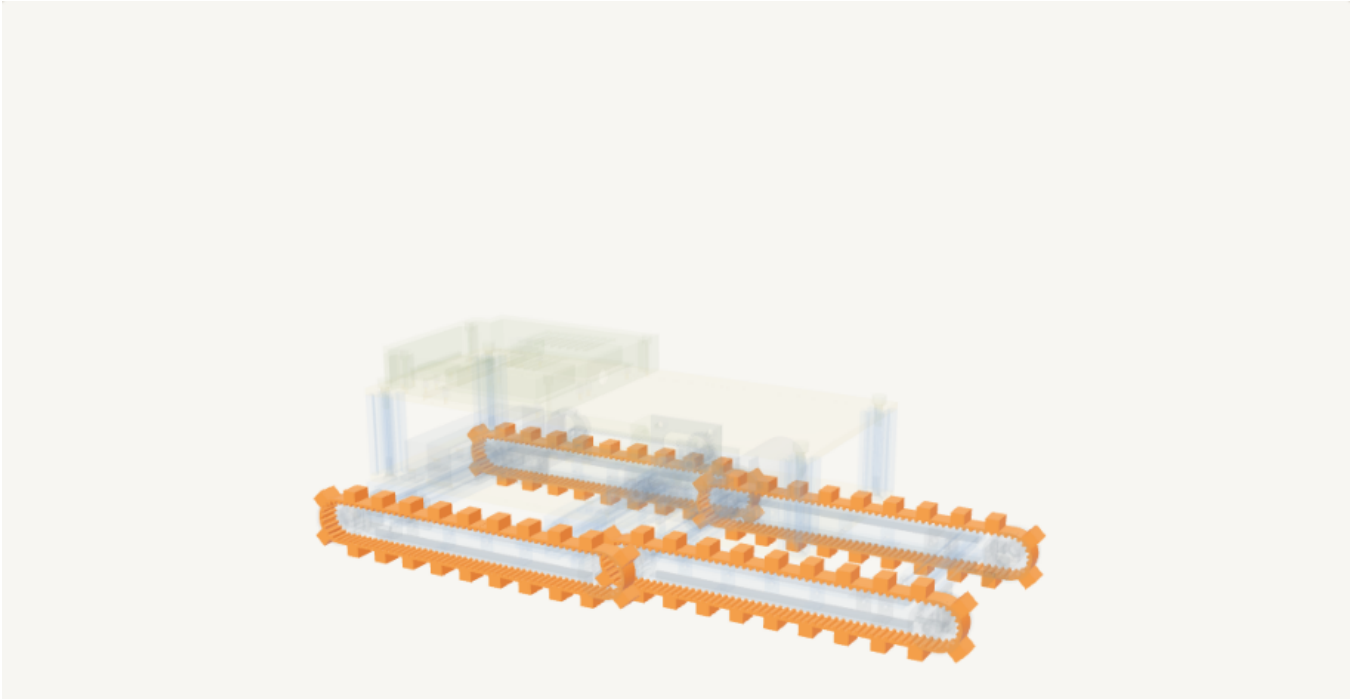
Where to get it

Fab Lab E4-02-06, Tiertime UP300 with the TPU head. File stl/track_TPU_x4.stl, standing as a ring, no supports, 0.2 mm, 100% infill. Filament: Fab Lab stock or Element14 SG TPU 95A (link). Drawing TS-01.

Link: Element14 SG: TPU 95A filament

https://sg.element14.com/multicomp-pro/mc011460/3d-printer-filament-tpu-1-75

Option B: Track belt HTD 550-5M-15 + 22 cleats (Rev B B10)



Quantity: 4 belts, 88 cleats

BUY belt; belt shop bonds the cleats

What it does

Nhien's stock-belt option on the same 20T wheels and 225 mm centres. The teeth inside drive; the bonded cleats grip the step edges. The cleat tips sit 25 mm above the hinge axis, level with the deck underside, so the deck needs a 13 x 135 notch over each front lane (or 5 mm cleats). The other version of this pack uses my printed track instead.

Where to get it

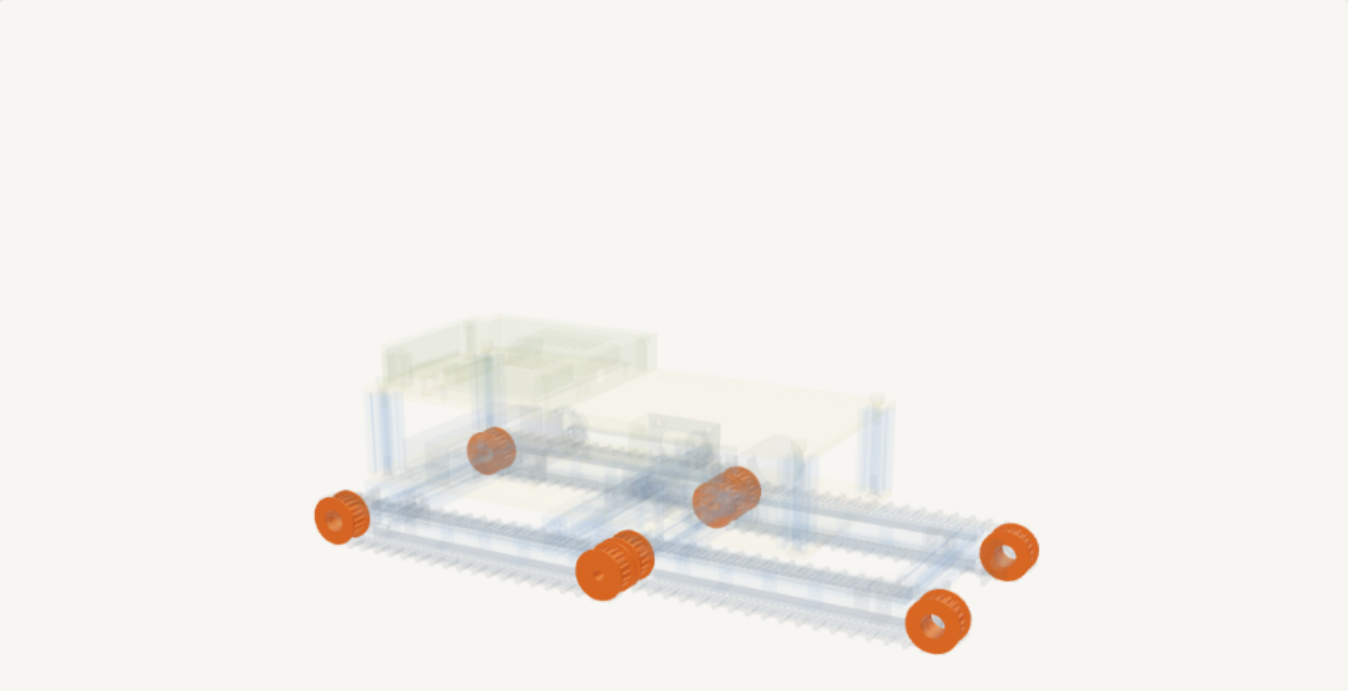
RS SG stock no. 058-6535 (SKF PHG 550-5M-15, link): 110 teeth, 15 wide. Cleats: rubber 10 x 7.9 x 15, one every 5th tooth, 22 per belt, cut and bonded by a belt shop (Fenner, Megadyne or Gates distributor). Drawing TS-01B.

Link: RS SG: search 058-6535

https://sg.rs-online.com/web/c/?searchTerm=058-6535

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Toothed wheel (timing pulley) HTD 5M 20T



Quantity: 8

BUY plain bore + MODIFY

What it does

Rev B's lanes are only 19-20 mm wide, so there is no room for a set-screw hub. The hinge ones are pinned to the plain round shaft (one spring pin each, design review); the end ones spin on bearings on a shoulder screw.

Where to get it

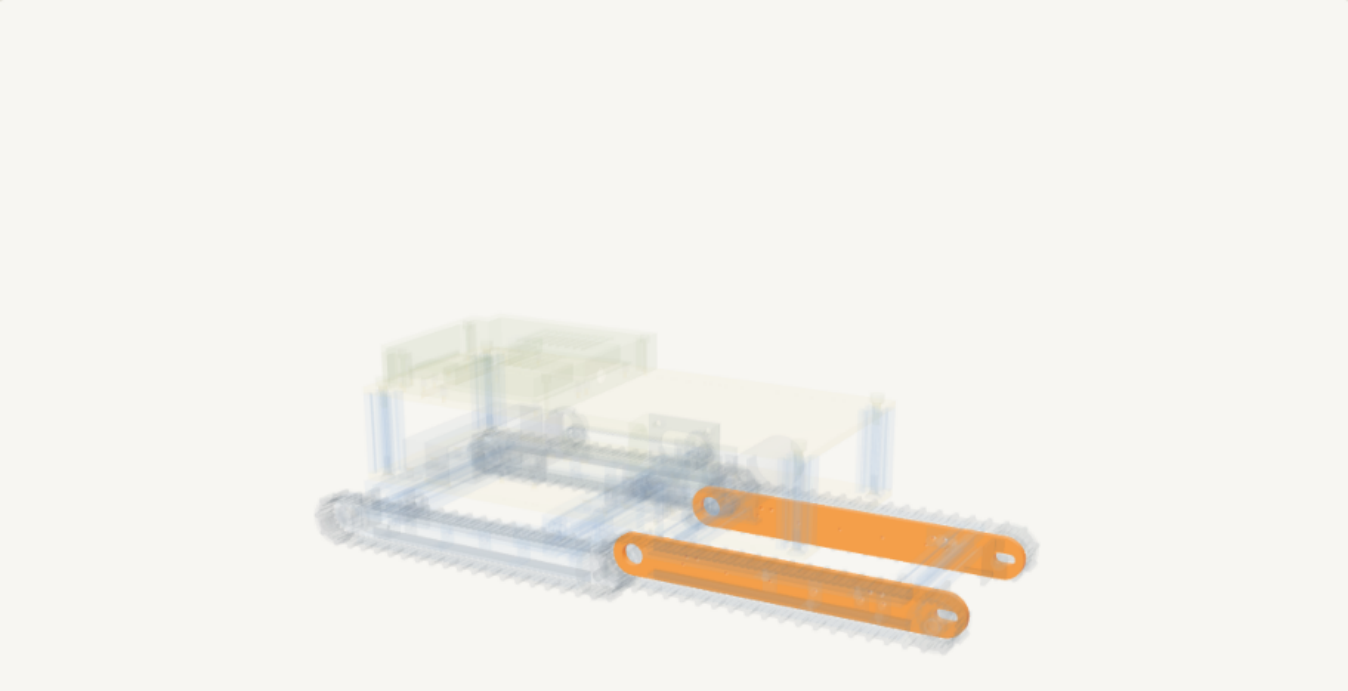
MISUMI SG, Timing Pulleys HTD 5M (link): 20 T, 15 mm belt, flanges both sides, PILOT (plain) bore. Then Professional Workshop: 4 reamed Ø8 H7 (hinge shaft, M1), 4 bored Ø16 H7 for two 688 bearings each (end idlers, M2). Cross holes Ø2.5 drilled with the shaft at assembly. TS-06, TS-11.

Link: MISUMI SG: HTD 5M timing pulleys

https://sg.misumi-ec.com/vona2/detail/110310532889/

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Side plate, front segment (Rev B P1)



Quantity: 2 (1 + 1 opposite hand)

MAKE: laser-cut aluminium (Nhien's)

What it does

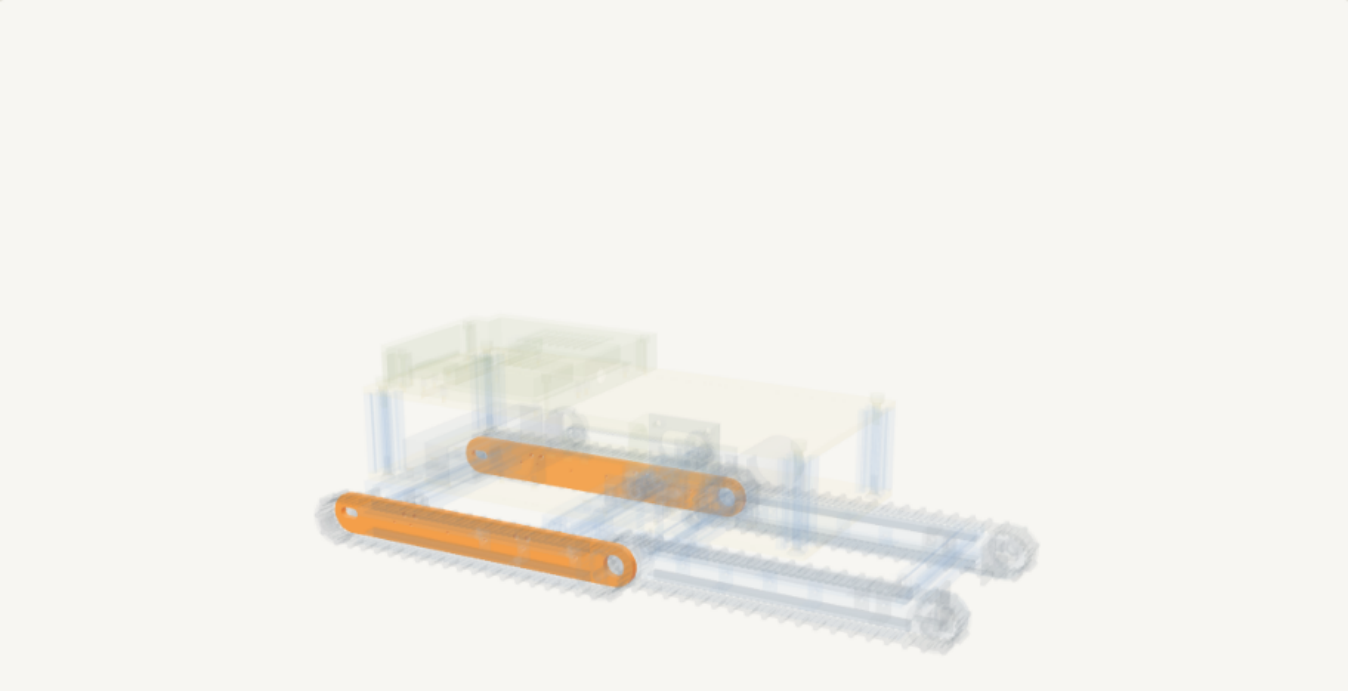
255 x 30.4 plate, inboard (z 55-60). Ø16 hole for the hinge bearing, a slot at the front end for the idler's shoulder screw (slide it to tension the track), tapped holes for the rail brackets and the 2020 bar brackets.

Where to get it

5 mm Al 6061, laser cut from DXF/P1 Side plate - FRONT segment.dxf, then drill and tap per Rev B sheet 4. Fab Lab laser does not cut aluminium: Professional Workshop or an outside laser shop on the vendor list.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Side plate, rear segment (Rev B P2)



Quantity: 2 (1 + 1 opposite hand)

MAKE: laser-cut aluminium (Nhien's)

What it does

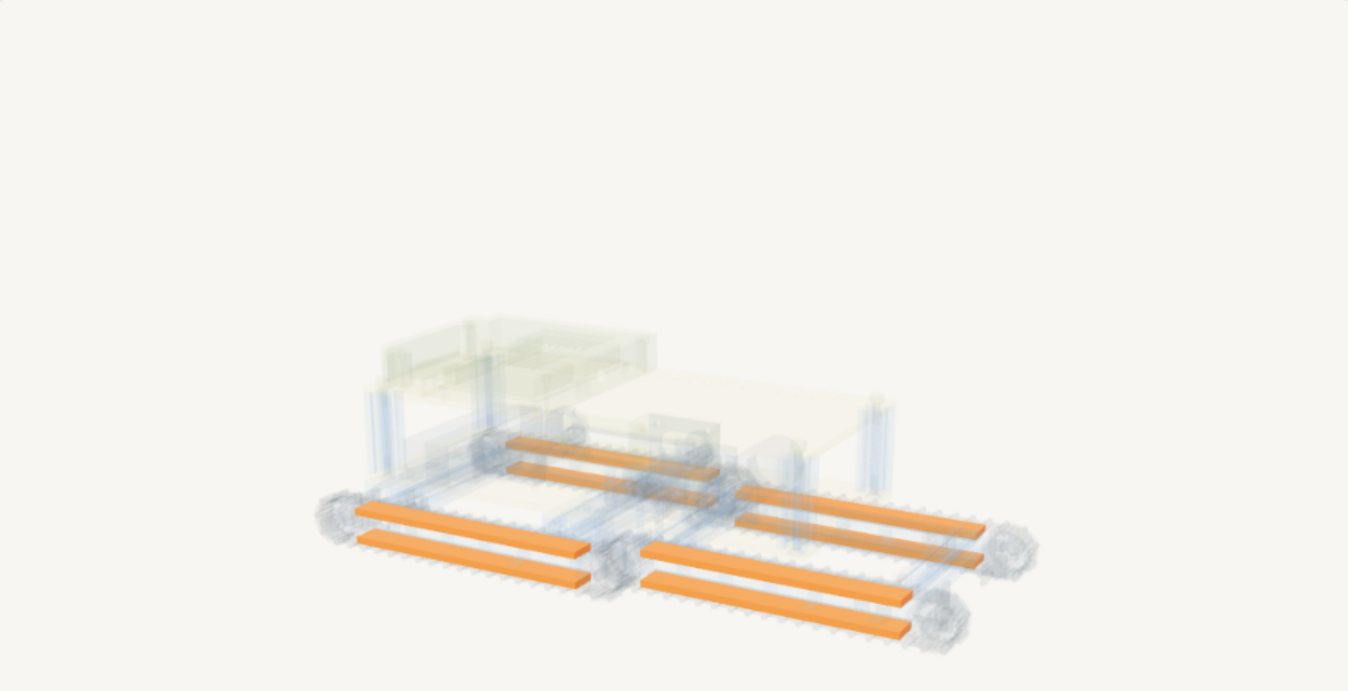
Outboard plate (z 80-85). The rear track runs outside it on a shoulder screw. Holes for the 3030 bar brackets and the rails.

Where to get it

As P1, from DXF/P2 Side plate - REAR segment.dxf, Rev B sheet 5.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Slide rails (Rev B P3-P6)



Quantity: 8

BUY strip + drill, trim to 186

What it does

The strips the track's upper and lower runs slide on so the track cannot sag between the wheels. The track's tooth tips ride on the lower one.

Where to get it

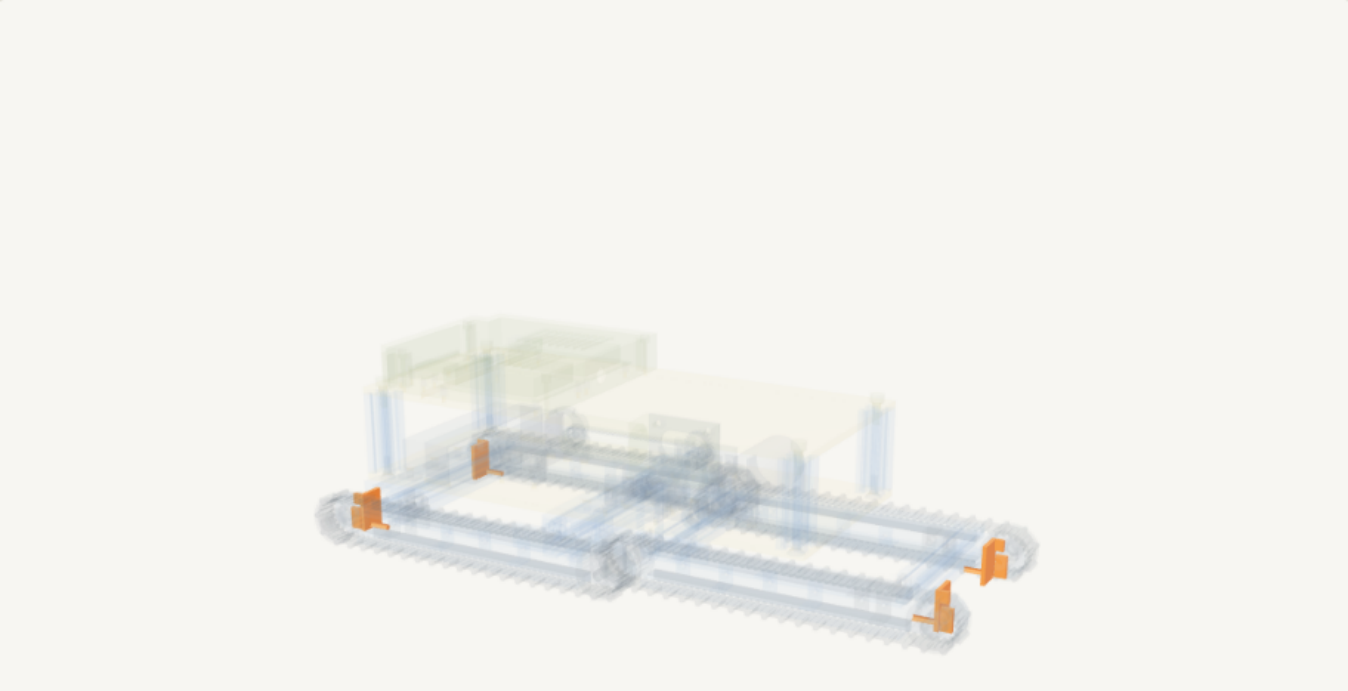
MISUMI SG UHMW-PE strip UPA-189-20-5 (link), 189 x 20 x 5, 3 pilot holes each, Rev B sheet 18. Trimmed to 186 at the idler end and carried on a full-length angle (TS-10) instead of the four RS clips.

Link: MISUMI SG: plastic strips (UHMW)

https://sg.misumi-ec.com/vona2/mech/M1500000000/M1501000000/

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Idler tensioner: tab, slider, jack screw (TS-09)



Quantity: 4 sets

MAKE on site

What it does

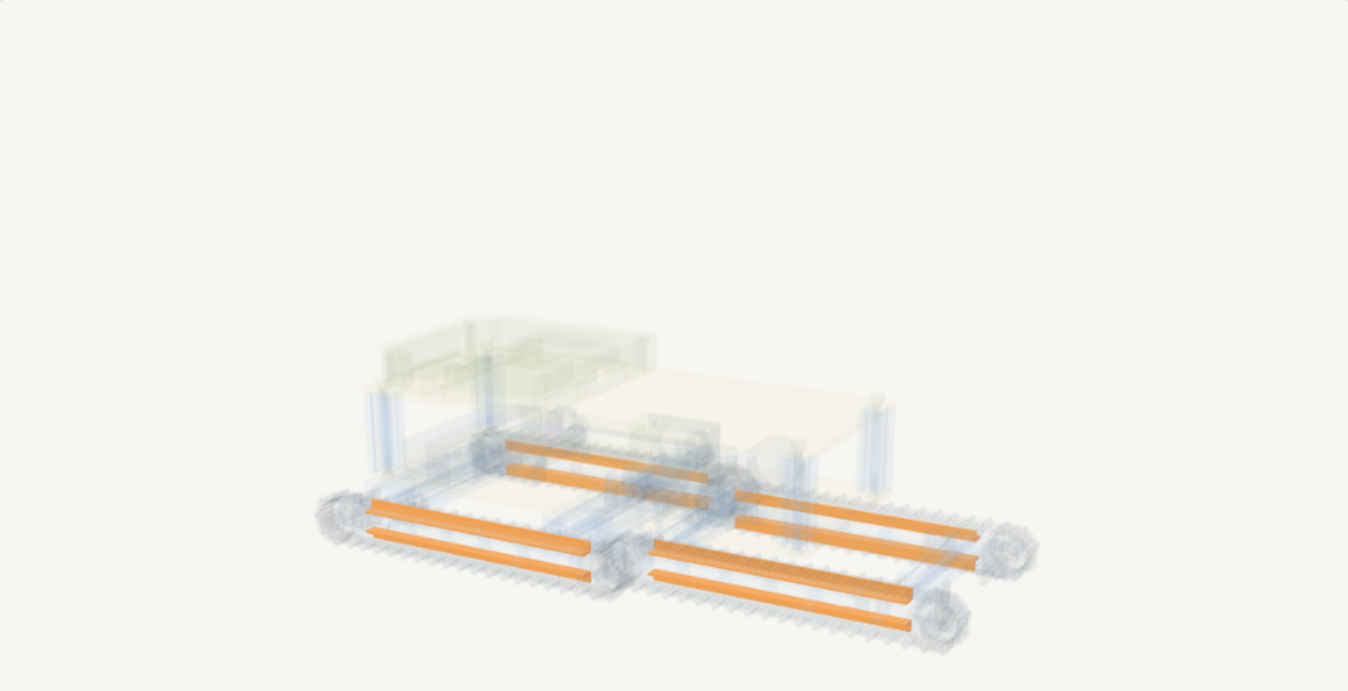
Gives the track a set tension. The jack screw pushes the idler outward along the plate slot until the upper run deflects 2 to 3 mm under a firm push, then the nyloc is tightened. Before this the shoulder screw was held by friction only.

Where to get it

Angle 20 x 20 x 3 cut 28 with an 8 x 6 ear and an M4 tapped hole; a 3 mm aluminium slider 8 x 16 with a Ø6.4 hole; an M4 x 20 jack screw; one M3 x 6 into a new tapped hole in the plate at (223, +10).

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Rail carrier angle (TS-10)



Quantity: 8

MAKE from angle stock

What it does

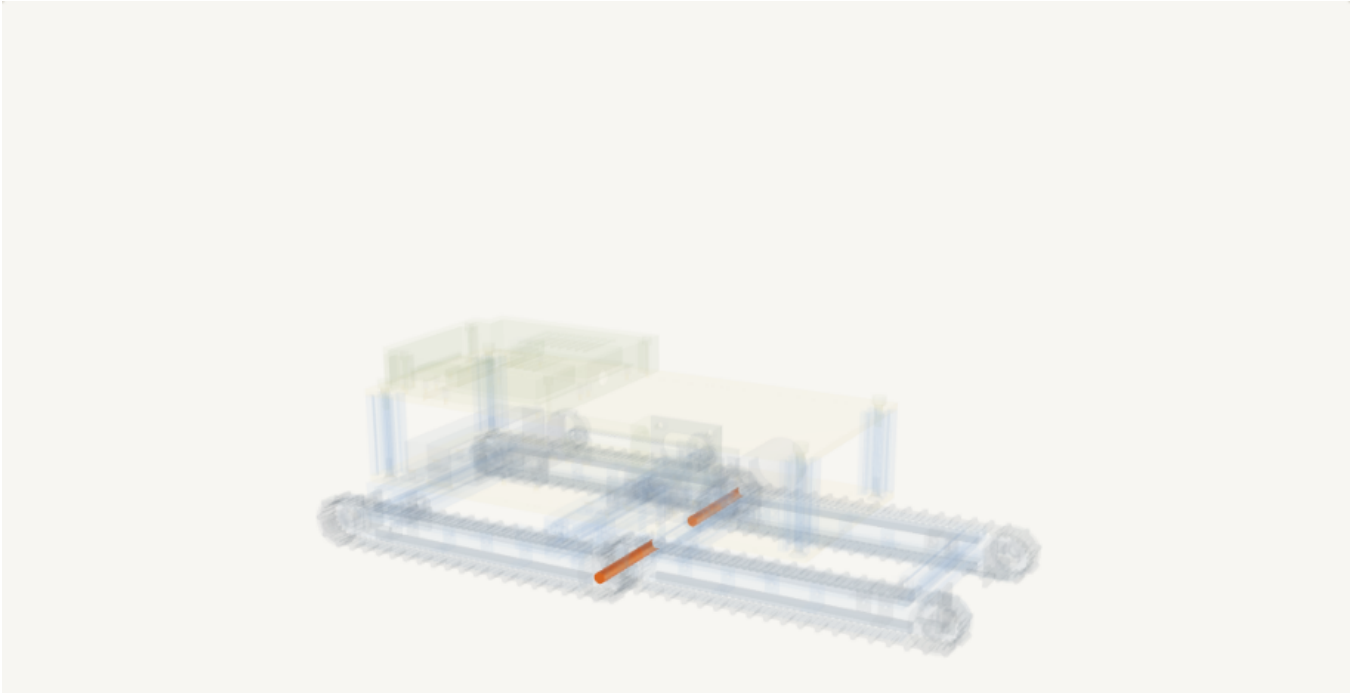
Replaces the four RS 427-991 clips per rail with 186 mm of contact on the plate and on the rail (design review: more bracket contact). Sits on the rail face away from the track.

Where to get it

Aluminium angle 10 x 10 x 1.5, 186 long, one leg cut to 8; 3 x Ø3.4 for the plate screws at the existing clip holes, 3 x Ø2.6 for the thread-forming rail screws. About 1.6 m of angle.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Hinge shaft (Rev B P20 A + B, now plain round)



Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Quantity: 2 (or 1 if one motor)

BUY round shaft, cut

What it does

The hinge axis and the drive. Runs in 688 bearings (reamed Ø16 H7 seats, Loctite 641) in both plates and in the P17 support. Located along its axis by the pinned front hinge pulley between the two plate bearings; three Ø2.5 spring pins per shaft.

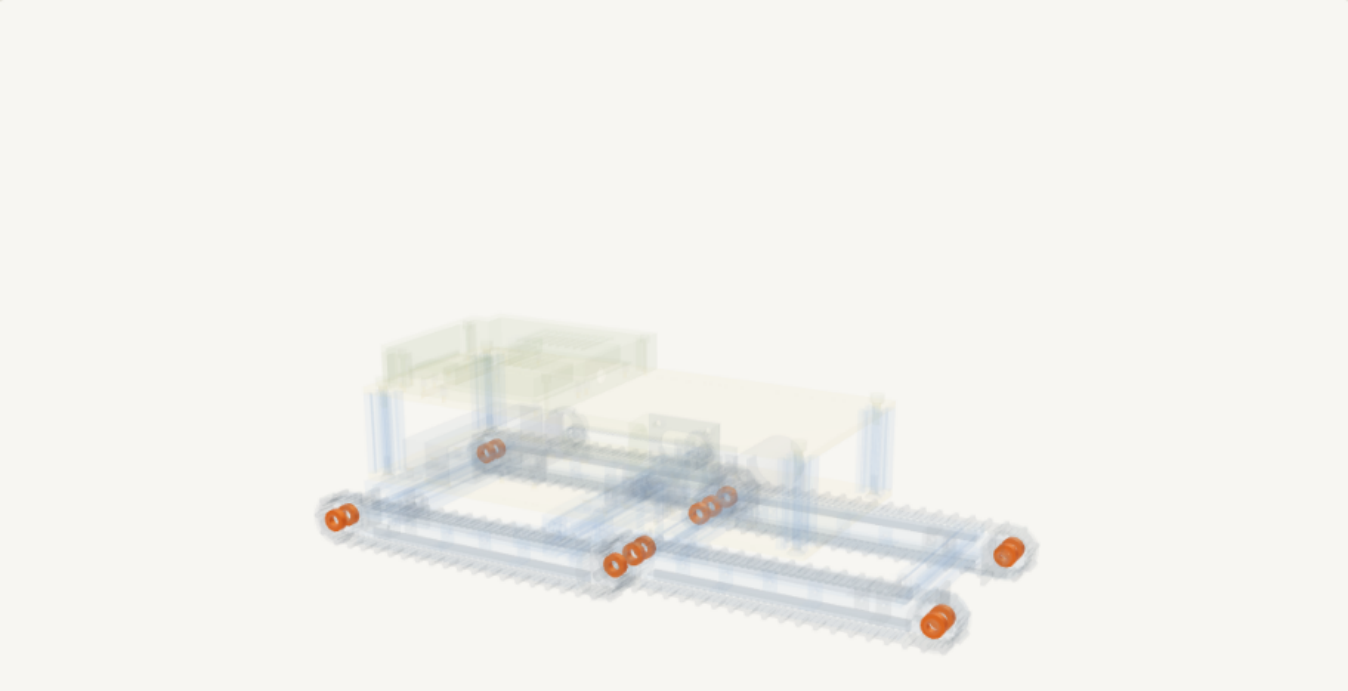
Where to get it

Ø8 h7 ground round shaft, 200 mm stock (MISUMI SG PSFJ8 or RS SG linear shafts, link). Cut to 82.3 (A) and 77.3 (B), chamfer. No D-flat and no E-clip grooves after the design review: the pulleys are pinned (TS-11). One motor per side -> two half shafts.

Link: RS SG: linear shafts

https://sg.rs-online.com/web/c/bearings-seals/linear-motion/linear-shafts/

Ball bearing 688-2RS 8 x 16 x 5



Quantity: 14

BUY: RS Singapore

What it does

Small OD (16) because the plate is only 30 mm tall. Pressed in first, before anything else goes together (build step 1).

Where to get it

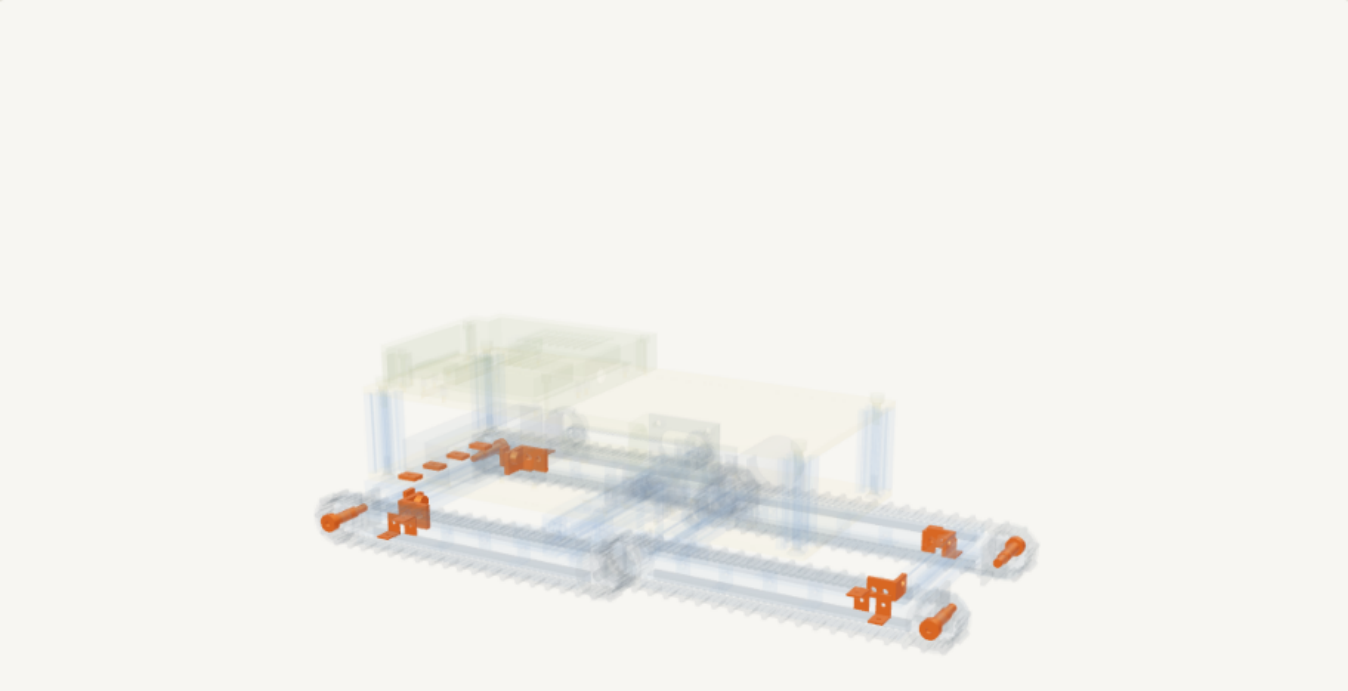
RS SG ball bearings (link): search 688-2RS. 4 in the plate hinge holes, 2 in the P17 supports, 8 inside the four idler pulleys.

Link: RS SG: ball bearings

https://sg.rs-online.com/web/c/bearings-seals/bearings/ball-bearings/

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Idler stub axle: shoulder screw Ø8 x 25, M6



Quantity: 4

BUY: MISUMI SG or RS SG

What it does

One per plate end, through the slot; the end wheel spins on two 688s on its shoulder. Slide outwards in the slot and tighten the nut to tension the track.

Where to get it

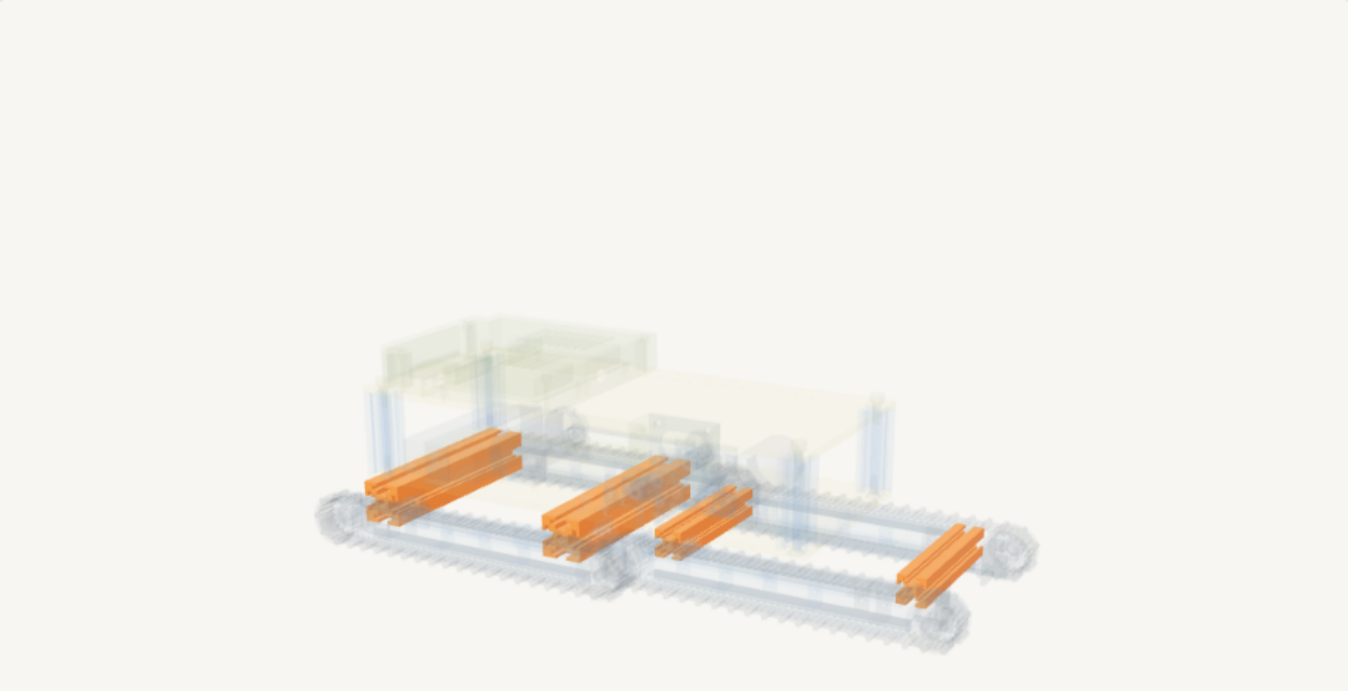
ISO 7379 8 x 25 shoulder screw (link), with 8 x 11 x 1 washer, 8 x 11 x 0.5 shim, 8 x 10 x 9 spacer, M6 washer and nyloc. Rev B sheet 3, idlers.

Link: RS SG: shoulder screws

https://sg.rs-online.com/web/c/fasteners-fixings/screws-bolts/shoulder-screws/

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Cross bars: 2020 x 110 and 3030 x 160 (Rev B P7, P8)



Quantity: 2 + 2

BUY profile, cut

What it does

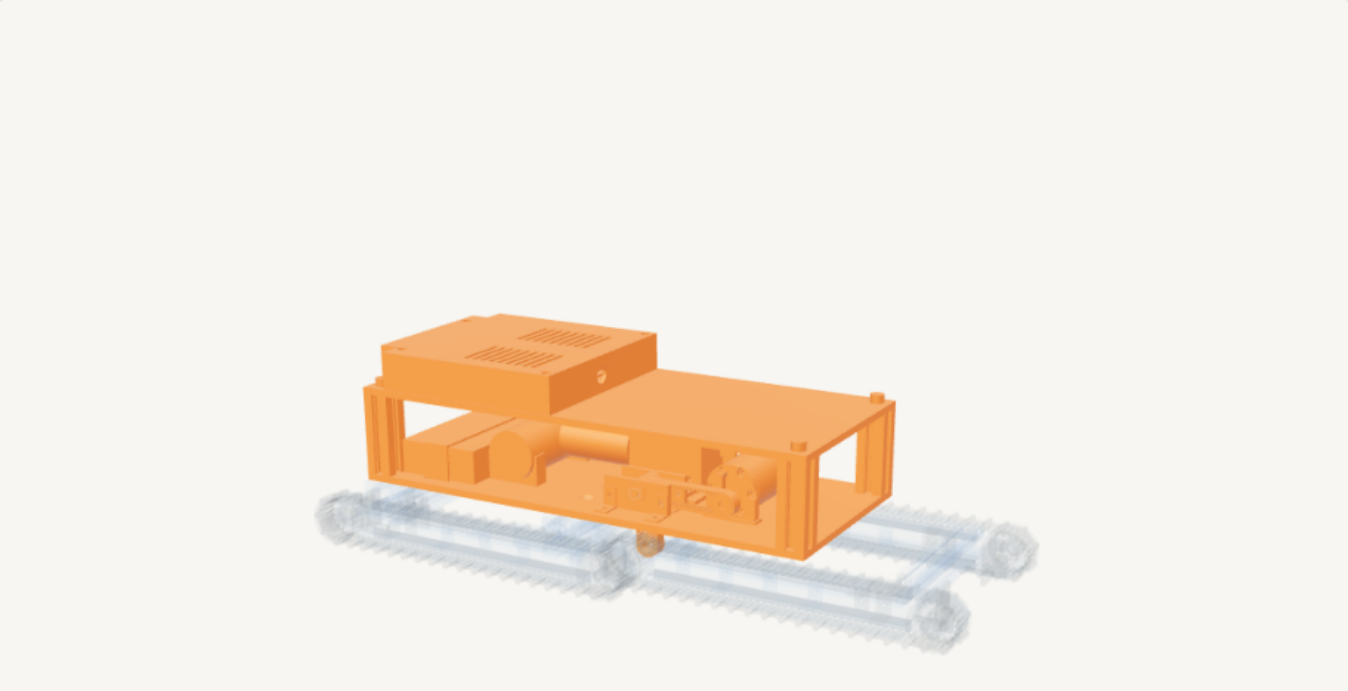
They hold the plates apart and carry the deck. The 2020 bars belong to the front segment and swing with it; the 3030 bars are what the deck bolts to. The near 2020 bar also carries the sector gear angle P14 and rests on the foam up-stops at 0°.

Where to get it

2020 B-type slot 6 (1 m) and 3030 slot 8 (0.5 m) from MISUMI SG or a local supplier; cut per Rev B sheet 19. Corner brackets P18/P19 from aluminium angle, T-nuts M5/M6.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Deck and tray (Rev B P12, P16) and everything on them



Quantity: 1 + 1

Nhien's parts

What it does

The 'box': 160 mm wide between the rear plates. Motors, drivers, battery and the platform mount sit here. Its edges overhang the front track lanes by 12 mm at 25 mm above the axis; slot H lets the sector gear through.

Where to get it

Al 5 mm laser cut, Rev B sheets 6-8; posts P15, controller cover P22, motors, electronics per the Rev B BOM.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.